BPL_TEST2_Fedbatch - demo

In [1]:
```
# Import the application model and context variables
from BPL_TEST2_Fedbatch_pyfmi import*
```

```
Windows - run FMU pre-compiled JModelica 2.14
```

In [2]:
```
# Import the general FMU_explore functions ana adapt them to the application
from fmu_explore_pyfmi import AdaptWith

Application = AdaptWith(parValue, parLocation, parCheck, \
                        model, fmu_model, fmu_process_diagram, \
                        MSL_usage, MSL_version, BPL_version, \
                        opts_std, simulationTime, timeDiscreteStates, \
                        diagrams, ax, lines)

par = Application.par
init = Application.init
disp = Application.disp
simu = Application.simu
show = Application.show

resetPen = Application.resetPen
process_diagram = Application.process_diagram
describe_general = Application.describe_general
describe_parts = Application.describe_parts
describe_MSL = Application.describe_MSL
FMU_explore_info = Application.FMU_explore_info
system_info = Application.system_info
SDG = Application.SDG
```

Bring in the user defined FMU_explore functions describe() and newplot()

In [3]:
```
run -i BPL_TEST2_Fedbatch_explore.py
```

```
Model for the process has been setup. Key commands:
 - par()       - change of parameters and initial values
 - init()      - change initial values only
 - simu()      - simulate and plot
 - newplot()   - make a new plot
 - show()      - show plot from previous simulation
 - disp()      - display parameters and initial values from the last simulation
 - describe()  - describe culture, broth, parameters, variables with values/units

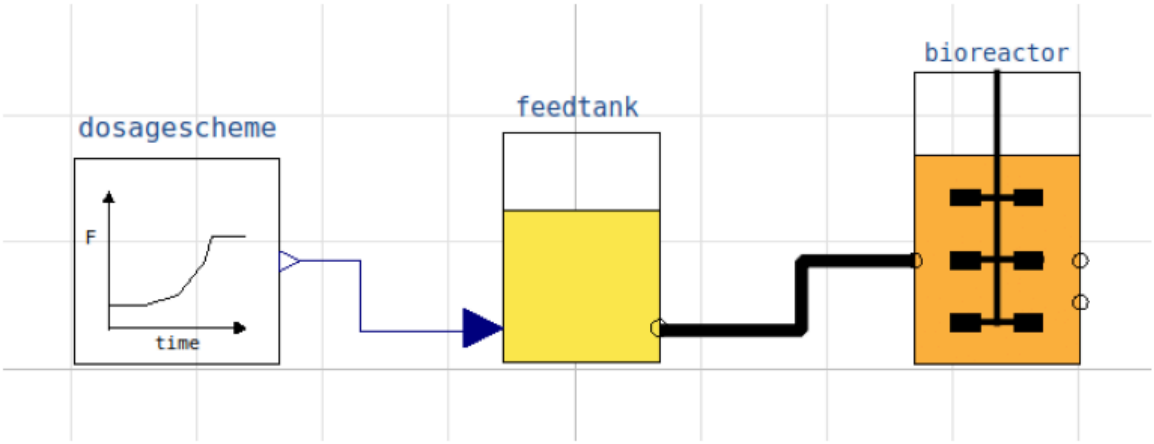
Note that both disp() and describe() takes values from the last simulation
and the command process_diagram() brings up the main configuration

Brief information about a command by help(), eg help(simu)
Key system information is listed with the command system_info()
```

In [4]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

In [5]:
```
process_diagram()
```

```
No processDiagram.png file in the FMU, but try the file on disk.
```



In [6]:
```
describe('process')
```

In [7]:
```
describe('culture')
```

```
Simplified text book model - only substrate S and cell concentration X
```

In [8]:
```
disp('dosagescheme')
```

```
F_start : 0.0
mu_feed : 0.2
t_startExp : 2.0
F_startExp : 0.12
F_max : 3.0
```

In [9]:
```
disp('feedtank')
```

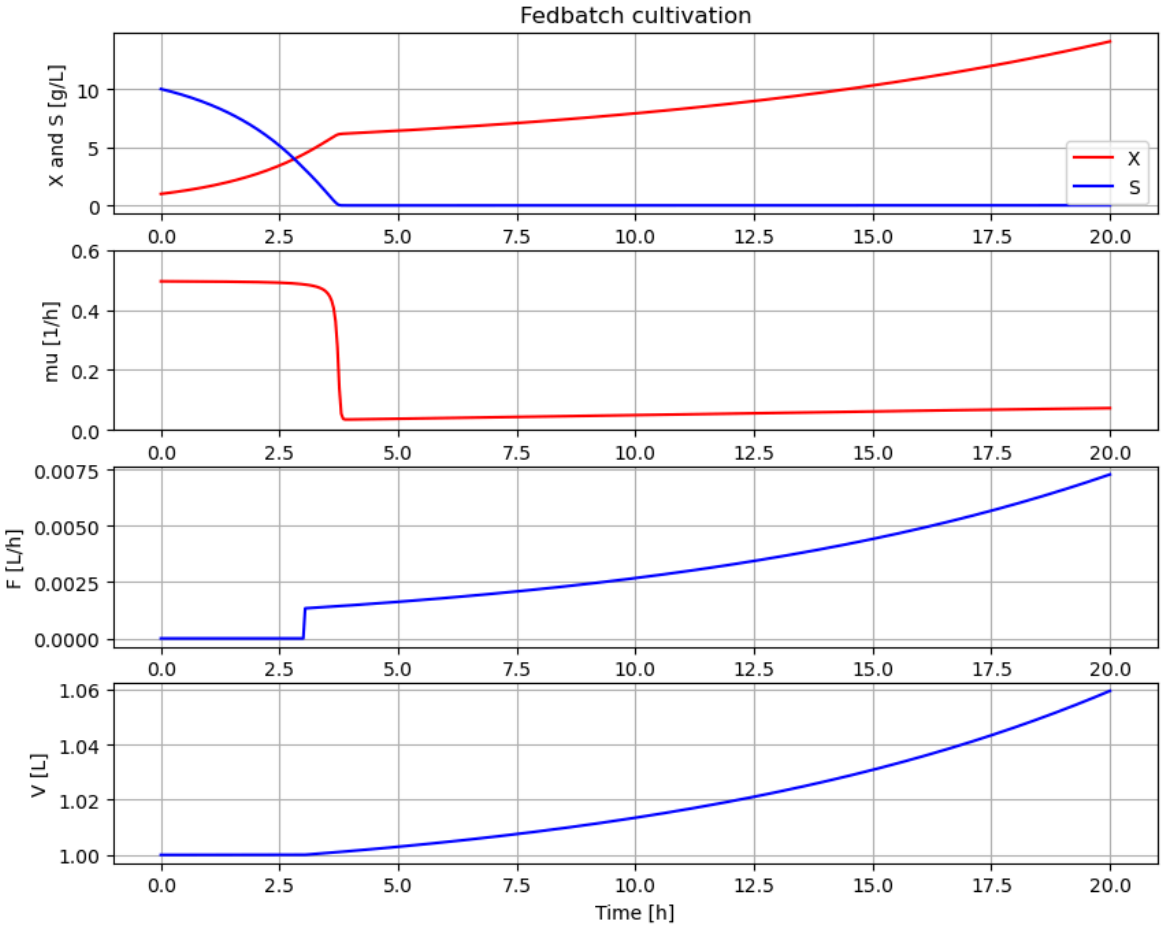
```
feedtank.S_in : 0.0
feedtank.V_start : 100.0
```

In [10]:
```
disp('bioreactor', mode='long')
```

```
bioreactor.V_start : V_start : 1.0
bioreactor.m_start[1] : VX_start : 0.0
bioreactor.m_start[2] : VS_start : 0.0
bioreactor.culture.Y : Y : 0.5
bioreactor.culture.qSmax : qSmax : 1.0
bioreactor.culture.Ks : Ks : 0.1
```

In [11]:
```
# Simulation with default values of the process
newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

simu(20)
```

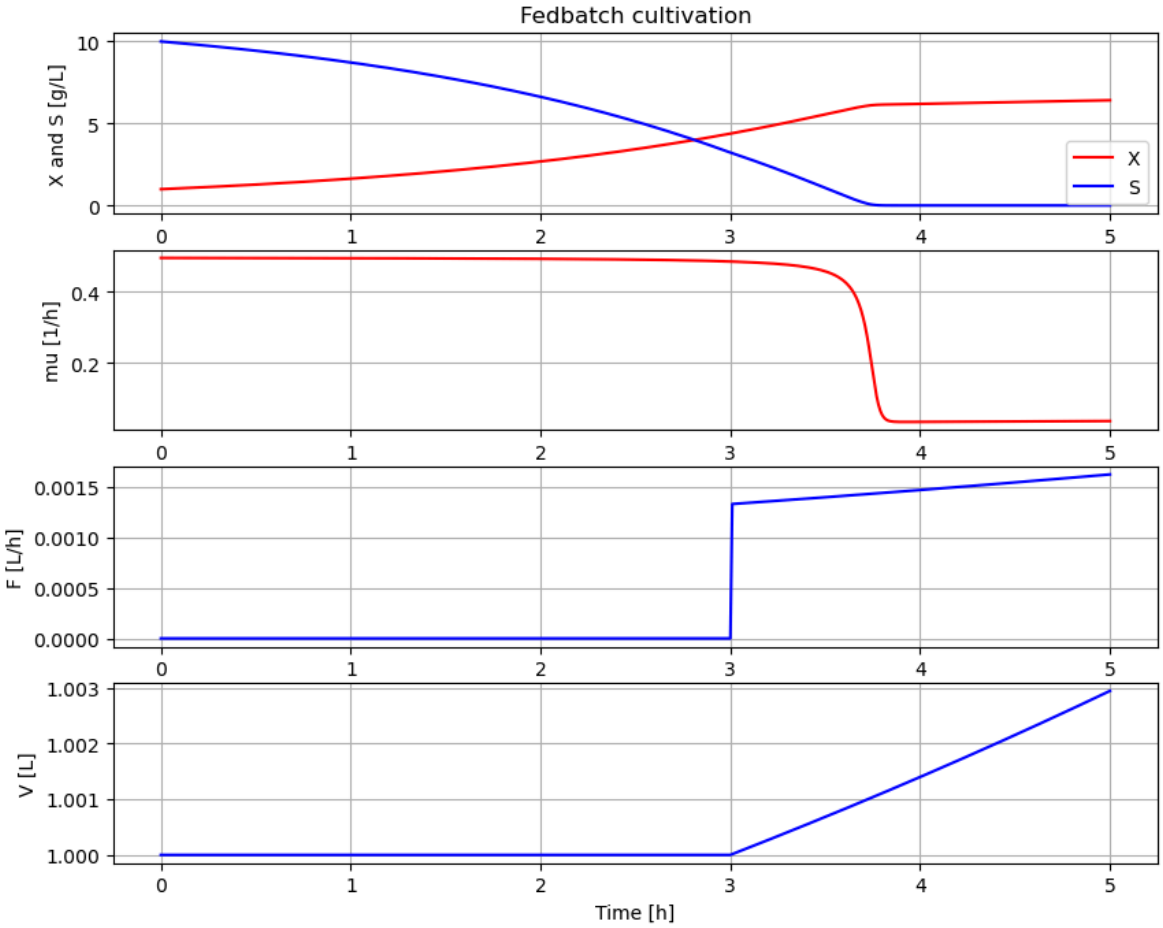


In [12]:
```
show()
```

In [13]:
```
simu()
```

In [14]:
```
show()
```

In [15]:
```
newplot()
show()
```



In [16]:
```
# Let us display and then save the feedprofile
disp('dosagescheme')
```

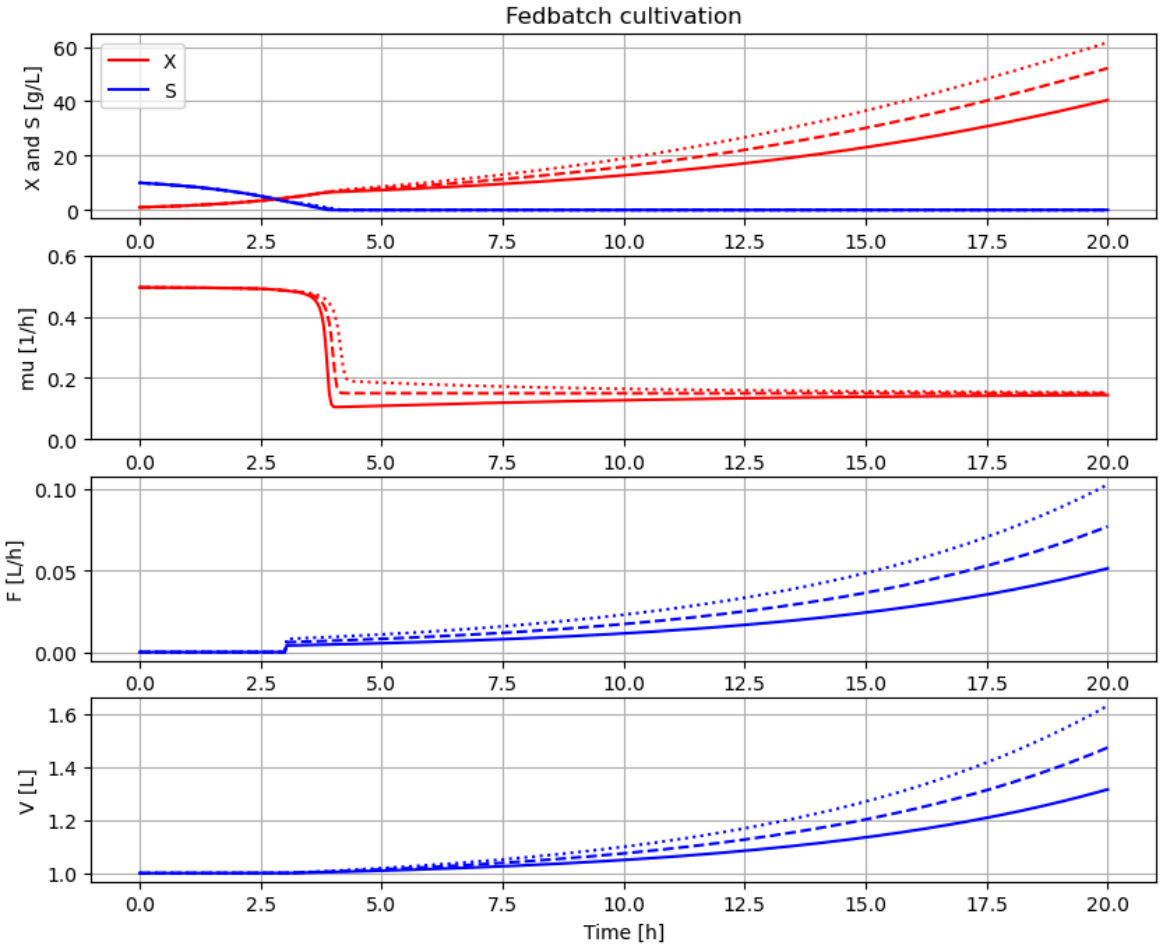
```
F_start : 0.0
mu_feed : 0.1
t_startExp : 3.0
F_startExp : 0.001
F_max : 0.3
```

In [17]:
```
feedprofile_1={'F_start':0, 'mu_feed':0.1, 't_startExp':3, 'F_startExp':0.001, '
```

In [18]:
```
# Let us develop a feedprofile with mu_feed=0.15 1/h instead
newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

# Choose mu_feed
par(mu_feed=0.15)

# Tune F_startExp value to get a stable culture growth rate at mu 0.15 1/h
for value in [0.004, 0.006, 0.008]:
    par(F_startExp=value)
    simu(20)
```

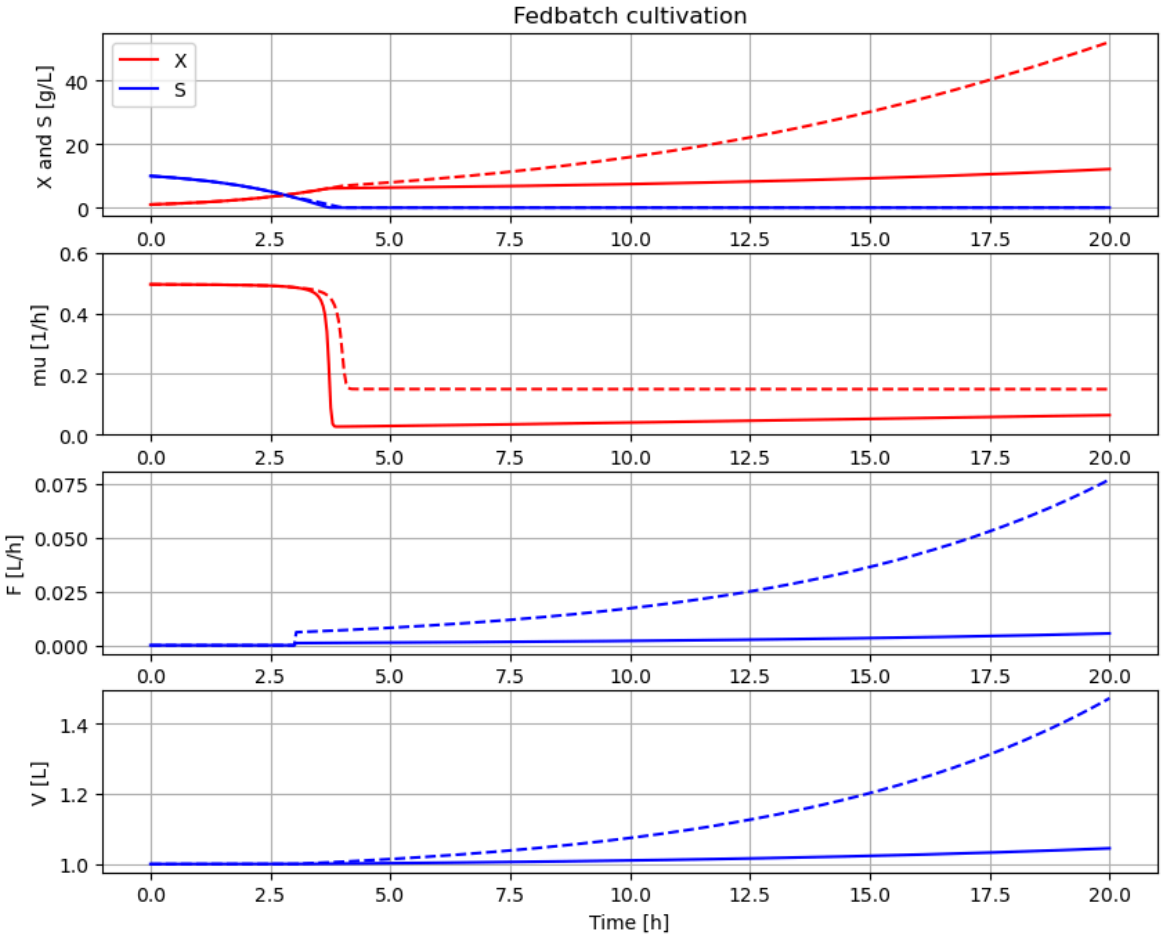


In [19]:
```
# We choose F_startExp = 0.008 and save the new feedprofile
feedprofile_2={'F_start':0, 'mu_feed':0.15, 't_startExp':3, 'F_startExp':0.006,
```

In [20]:
```
# Compare the result of ot the two
newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

for feedprofile in [feedprofile_1, feedprofile_2]:
    par(feedprofile)
    simu(20)

# Restore the original feedprofile
par(feedprofile_2)
```



In [21]:
```
describe('mu')
```

```
Cell specific growth rate variable : 0.15 [ 1/h ]
```

In [22]:
```
describe('parts')
```

```
['bioreactor', 'bioreactor.culture', 'dosagescheme', 'feedtank', 'MSL']
```

In [23]:
```
describe('MSL')
```

```
MSL: RealInput, RealOutput
```

In [24]:
```
system_info()
```

```
System information
 -OS: Windows
 -Python: 3.12.11
 -Scipy: 1.16.0
 -PyFMI: 2.22.0
 -FMU by: JModelica.org
 -FMI: 2.0
 -Type: FMUModelCS2
 -Name: BPL.Examples_TEST2.Fedbatch
 -Generated: 2025-07-26T09:38:43
 -MSL: 3.2.2 build 3
 -Description: Bioprocess Library version 2.3.1
 -Interaction: FMU_explore version 1.2.4
```

In [25]:
```
SDG()
```